

Adaptive Prompt Sensitivity in Complex LLM Tasks

Research Methodology Report

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Course / Project: Prompt Sensitivity Research Pipeline

Abstract. This report presents the complete methodology of an end-to-end research pipeline that investigates and mitigates **prompt sensitivity** in large language models — the phenomenon where semantically equivalent but lexically different prompts produce inconsistent outputs. The pipeline spans four phases: (1) benchmark dataset generation using GenSens, (2) quantitative robustness evaluation via the PRI Benchmark, (3) an inference-time mitigation technique called LL-PIRC (Logit-Lens Paraphrase-Invariant Residual Clamping), and (4) rigorous statistical validation.

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1. Problem Statement & Motivation

Large language models (LLMs) are increasingly deployed in real-world applications that depend on reliable, consistent outputs. However, a well-documented phenomenon called **prompt sensitivity** causes LLMs to produce qualitatively different responses when presented with semantically equivalent but lexically varied prompts. For example, 'Summarise the article in three sentences.' and 'Provide a brief three-sentence summary of the article.' should elicit nearly identical responses — yet in practice they often do not.

Why This Matters

- **Reproducibility:** scientific or engineering workflows require consistent model behaviour across prompt phrasings.
- **Fairness:** users who phrase the same query differently should not receive systematically worse answers.
- **Reliability:** downstream pipelines built on LLM outputs fail unpredictably when sensitivity is high.
- **Evaluation validity:** benchmarks that use a single prompt template may over- or under-estimate true model capability.

Research Questions

- RQ1: How can prompt sensitivity be quantified rigorously across multiple semantic axes?
- RQ2: Is it possible to detect the model-internal layer responsible for sensitivity?
- RQ3: Can an inference-time intervention reduce sensitivity without retraining the model?
- RQ4: Do the sensitivity reduction gains hold up under rigorous statistical testing?

Scope. The pipeline operates on four task domains: **Summarization** (CNN/DailyMail news articles), **Creative generation** (article expansion from highlights), **Dialogue understanding** (DREAM multi-turn QA), and **Explanatory QA** (ELI5 question–answer pairs). Evaluation is conducted on 3×7–8B instruction-tuned models and one 70B AWQ-quantised model.

2. Pipeline Overview

The pipeline consists of four sequential phases. Each phase produces durable artefacts (JSONL datasets, CSV result files) that downstream phases consume, enabling crash-safe, resumable execution and separation of expensive GPU steps.

Phase	Name	Primary Output	Key Models
Phase 1	GenSens Dataset Generation	gensens__Ninst_Kvar.jsonl	LLaMA-3.1-70B-AWQ (generator) SBERT all-mpnet-base-v2 (filter)
Phase 2	PRI Benchmark Evaluation	responses.csv scored_samples.csv	3×8B + 1×70B subject models BAAI/bge-large-en-v1.5 (embedder) LLaMA-3.1-70B (judge/NLI)
Phase 3	LL-PIRC Intervention	pirc.jsonl / pirc.json	LLaMA-3.1-8B-Instruct (HF hooks required)
Phase 4	Statistical Validation	eval_summary.json plots/	CPU-only (scipy, numpy)

3. Phase 1 — GenSens: Benchmark Dataset Generation

GenSens is a controlled benchmark dataset generator that produces sets of **semantically equivalent but lexically diverse** prompt paraphrases. Each dataset instance consists of one base prompt and K=8 filtered paraphrase variants, all paired with a gold reference output.

3.1 Task Domains

Task	Source Dataset	What is Paraphrased	Gold Reference
Summarization	CNN/DailyMail 3.0.0	Summarisation instruction	Highlights
Creative	CNN/DailyMail 3.0.0	News highlights (premise)	Full article
Dialogue	DREAM	Comprehension question	Correct answer choice
QA	ELI5 (sentence-transformers)	Explanatory question	Answer/explanation

3.2 Paraphrase Generation Strategy

The generator employs LLaMA-3.1-70B-Instruct-AWQ-INT4 (via vLLM) with **16 distinct paraphrase strategies**, including: synonym substitution, passive/active voice transformation, sentence reordering, role-prefix injection, hedged phrasing, formality shift, and question-form conversion. A **per-task strategy whitelist** restricts strategies to those that preserve intent for each task domain.

3.3 Multi-Stage Quality Filtering

- **SBERT similarity band:** Candidates must satisfy $0.82 \leq \text{cosine}(\text{base}, \text{candidate}) \leq 0.98$ using all-mpnet-base-v2. The upper bound rejects trivial restatements; the lower bound rejects meaning-drift.
- **Lexical divergence floor:** Token Jaccard overlap between candidate and base (or any already-accepted variant) must be ≤ 0.85 to ensure lexical diversity.
- **Per-strategy cap:** At most 2 variants per strategy family, ensuring the K=8 set spans multiple strategy types.
- **Bidirectional NLI gate (QA/Dialogue only):** Both $P(\text{entail} \mid \text{base} \rightarrow \text{candidate}) \geq 0.50$ AND $P(\text{entail} \mid \text{candidate} \rightarrow \text{base}) \geq 0.50$, verified with cross-encoder/nli-deberta-v3-small. Prevents passive–active swaps that change the answer.
- **Per-task SBERT threshold:** Dialogue uses a relaxed threshold of 0.78 (vs. default 0.82) to recover variant density for short conversational utterances.

3.4 Summarization Instruction Pool

For summarisation tasks, a pool of 100 unique instructions is pre-generated using Qwen2.5-7B-Instruct across four complexity levels (simple ≤ 12 words $\rightarrow$ complex 35–60 words). Each dataset instance is

assigned a unique instruction (level-sorted), creating a complexity gradient that tests model sensitivity to instruction elaboration.

3.5 Adversarial Paraphrase Subset

Five deterministic perturbation families generate an adversarial subset to stress-test robustness beyond easy LLM rewrites: **typo** (8% character swaps), **sentence reorder**, **double negation** ($\neg\neg X \equiv X$), **hedged phrasing**, and **formality shift**. PRI is reported separately on the easy GenSens subset and the adversarial subset.

3.6 Dataset Scale

- Target: 200 instances $\times$ 4 tasks = 800 samples
- K = 8 paraphrase variants per instance
- All instances carry a non-empty gold reference_output
- Output format: JSONL with per-variant strategy tags, SBERT scores, and NLI audit fields

4. Phase 2 — PRI Benchmark: Robustness Evaluation

The PRI (Prompt Robustness Index) Benchmark evaluates how consistently each subject model responds across the K=8 paraphrase variants of every prompt. The pipeline is divided into a **generate-only** phase (one subject model at a time) and a **score-from-CSV** phase (embedder + judge), ensuring no two large models co-reside on the same 80 GB GPU card.

4.1 Subject Models Under Evaluation

- meta-llama/Llama-3.1-8B-Instruct
- Qwen/Qwen2.5-7B-Instruct
- mistralai/Mistral-7B-Instruct-v0.3
- hugging-quants/Meta-Llama-3.1-70B-Instruct-AWQ-INT4 (AWQ Marlin quantisation)

All models use greedy decoding (do_sample=False, seed=42) for reproducibility. Chat-template formatting is applied for all instruct-tuned models.

4.2 Core Metric Suite

Metric	Abbreviation	What It Measures	Range
Semantic Manifold Stability	SMS	Mean cosine similarity minus embedding variance penalty	↑ [0,1]
Performance Elasticity	AUC-E	ROUGE-L stability across variant indices (AUC of performance curve)	↑ [0,1]
Thematic Robustness Drift	TRD	Embedding-space semantic drift across variants	↓ [0,1]
Key Point Info Gain	KPIG	Fraction of reference key-facts covered across variants	↑ [0,1]
PPL Variance	PPL_var	Perplexity variance across variants (intra-model diagnostic)	↓ [0,1]
Branching Factor	BF	Token-level entropy variance (intra-model diagnostic)	↓ [0,1]
Correctness Score	CS	0.40·semantic + 0.20·length_adequacy + 0.40·reference coverage	↑ [0,1]
Faithfulness	Faith.	NLI entailment P(entail)+0.5·P(neutral), source→response	↑ [0,1]

Hallucination Score	HS	Fraction of ungrounded content words (diagnostic only)	↓ [0,1]
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4.3 Composite Scores

Prompt Robustness Index (PRI) — Primary Ranking Score

$$\text{PRI} = 0.40 \times \text{SMS} + 0.35 \times \text{CS} + 0.25 \times \text{Faithfulness}$$

Three non-collinear axes: semantic consistency (SMS), reference quality (CS), and NLI-grounded faithfulness. A short-output gate ($\text{avg_len} < 12$ tokens) applies a $\times 0.85$ penalty.

Observable Robustness Index (ORI) — Cross-Model Ranking

$$\text{ORI} = (\text{SMS} + \text{AUC-E} + (1 - \text{TRD}) + \text{KPIG}) / 4$$

Arithmetic mean over the four external-consistency axes. Used for cross-model ranking tables.

Intrinsic Fidelity Index (IFI) — Intra-Model Diagnostic

$$\text{IFI} = 1 - (\text{PPL_var} + \text{BF}) / 2$$

Intra-model diagnostic measuring internal generation stability. PPL/BF are architecture-dependent and not cross-model comparable.

Final Score

$$\text{Final_Score} = 0.60 \times \text{PRI} + 0.40 \times \text{Human_Score}$$

4.4 Diagnostic Dual-Pillar Score (Publication Appendix)

A separate harmonic diagnostic stack is used for causal diagnosis and publication appendix tables. It is intentionally strict: if any pillar approaches zero, the composite also approaches zero.

$$\text{Diagnostic_ORI} = \text{WHM}(\text{SMS}, \text{AUC-E}, \text{KPIG}, 1-\text{TRD}) \text{ weights: } 0.40 / 0.30 / 0.20 / 0.10$$

$$\text{Diagnostic_IFI} = \text{WHM}(1-\text{PPL_var}, 1-\text{BF}, 1-\text{PC_stab}) \text{ weights: } 0.50 / 0.30 / 0.20$$

$$\text{Diagnostic_PRI} = \text{HM}(\text{Diagnostic_ORI}, \text{Diagnostic_IFI}) \text{ weights: } 0.50 / 0.50$$

The TRD weight is demoted to 0.10 because TRD and SMS have Pearson $r = -0.98$ (nearly collinear), so equal weighting would double-count consistency. The weighted harmonic mean (WHM) enforces the 'any zero $\rightarrow$ composite zero' property.

4.5 2×2 Diagnosis Matrix

	IFI ≥ 0.70 (Internally Stable)	IFI < 0.70 (Internally Fragile)
ORI ≥ 0.70	✓ ROBUST	Externally Stable / Internally Fragile
ORI < 0.70	Internally Stable / Output-Sensitive	✗ FRAGILE

4.6 Scoring Infrastructure

- **Embedder:** BAAI/bge-large-en-v1.5 (335M, MTEB ~64) — drives SMS, CS semantic component, Semantic TRD, and advanced KPIG.
- **LLM Judge:** LLaMA-3.1-70B-AWQ (independent of subjects) — provides Human_Score and Faithfulness-LLM-as-NLI.
- **Incremental CSV writing:** every row is fsync'd to disk immediately — crash-safe resume.
- **Disk caching:** MD5-keyed CacheManager avoids redundant model calls on resume.
- **IFI normalization:** PPL_var and BF are min-max normalised per model before computing IFI to prevent saturation artefacts.

5. Phase 3 — LL-PIRC: Inference-Time Mitigation

LL-PIRC (**Logit-Lens Paraphrase-Invariant Residual Clamping**) is a *training-free* inference-time intervention that reduces prompt sensitivity by steering the model's internal hidden states toward a cross-paraphrase consensus representation at the layer most responsible for sensitivity.

5.1 Step 1 — Logit Lens: Per-Layer PPL Extraction

For each paraphrase variant k and each transformer layer ℓ , the Logit Lens projects the intermediate hidden state $h^\ell(\ell)$ through the final LayerNorm and language-model head to obtain vocabulary logits. Per-token perplexity is then computed from these intermediate logits:

$$\begin{aligned} \text{logits}^\ell(\ell) &= \text{lm_head}(\text{LayerNorm}(h^\ell(\ell))) \\ \text{PPL}^\ell(\ell)_k &= \exp(-\text{mean}_t [\log P_t^\ell(\ell)]) \end{aligned}$$

Hidden states are cast to float32 for numerical stability during the Logit Lens computation.

5.2 Step 2 — Sensitive Layer Detection (ℓ^*)

For each layer ℓ , a sensitivity signal $S(\ell)$ measures how much the layer's intermediate PPL varies across the K paraphrase variants:

$$S(\ell) = \text{Var}_k [\text{mean_token_PPL at layer } \ell]$$

The **sensitive layer** ℓ^* is identified as the layer with the steepest upward jump in this curve:

$$\ell^* = \text{argmax}_\ell [S(\ell) - S(\ell-1)]$$

- Layer scan range: 25% to 100% of total transformer depth (early layers carry syntax, not prompt intent).
- Fallback (Z-score method): first ℓ where $S(\ell) > \text{mean}_S + 2.0 \times \text{std}_S$, if inflection detection fails.
- Ablation knob: `--ell-star L` forces ℓ^* to a fixed layer for sensitivity sweeps.

5.3 Step 3 — Anchor Token Identification

At layer ℓ^* , anchor tokens are the subset of input tokens whose hidden states are most stable across paraphrase variants — i.e., the tokens that reliably encode content regardless of surface phrasing.

$$\text{Anchor score}_t = \text{rank}(\text{mean_ppl}_t) + \text{rank}(\text{var_ppl}_t) \text{ [lower = more stable]}$$

- Selection: bottom 30% by anchor score (percentile-based, model-agnostic).
- Pre-filtering: punctuation and stopwords are excluded — only content words are anchor candidates.
- Guarantee: at least 1 anchor is always selected; at most $\ell_{\text{min_len}} / 2$ anchors.
- NaN/Inf handling: non-finite PPL positions are hard-excluded before ranking.

5.4 Step 4 — Residual Clamping

At inference time, a PyTorch forward hook is registered on transformer layer $\blacksquare^*$. The hook fires **only during the prefill pass** (the single forward pass over the full prompt), then disables itself for all subsequent decode steps:

$$\mathbf{h_clamped[anchors]} = \alpha \times \mathbf{mean_h[anchors]} + (1 - \alpha) \times \mathbf{h_original[anchors]}$$

- $\mathbf{mean_h[anchors]}$ = arithmetic mean of anchor hidden states across all K paraphrase variants.
- $\alpha = 1.0$ (default, full replacement): complete substitution with consensus representation.
- $\alpha < 1.0$ (soft clamping): convex interpolation preserving some prompt-specific signal.
- Prefill-only constraint: prevents degenerate autoregressive loops that occur when clamping fires on single-token decode steps.
- Ablation knob: `--alpha A` overrides α from the CLI for systematic sensitivity sweeps.

5.5 Evaluation Protocol

One PIRC-stabilised output is generated per paraphrase variant, yielding K outputs per article. ROUGE-L variance across these K outputs is the primary mitigation metric.

$$\mathbf{PIRC\ Variance} = \mathbf{Var(ROUGE-L\ scores\ across\ K\ PIRC\ outputs)}$$

- Fixed preregistered τ / α (`dev_tune_articles = 0`) to avoid oracle overfitting.
- Baseline: greedy generation without clamping, under identical prompt variants.
- No-training baselines for comparison: temperature smoothing, self-consistency voting, system-prompt stabilisation, in-context learning exemplars.

5.6 Supported Architectures

- LLaMA-style: forward hook on `model.model.layers[\blacksquare^*]`, final norm = `model.model.norm`.
- GPT-2-style: forward hook on `model.transformer.h[\blacksquare^*]`, final norm = `model.transformer.ln_f`.
- Hook handles both tuple and plain Tensor outputs (transformers v5.x compatibility).

6. Phase 4 — Statistical Evaluation & Validation

Phase 4 rigorously validates the LL-PIRC intervention using non-parametric hypothesis tests, bootstrap confidence intervals, and paired effect sizes.

6.1 Variance Reduction

Variance Reduction = $1 - \text{mean}(\text{var_pirc}) / \text{mean}(\text{var_baseline})$

Positive values indicate that PIRC outputs are more consistent across paraphrase variants than the unmodified baseline.

6.2 Hypothesis Tests

- **Wilcoxon Signed-Rank (Variance):** One-sided test — H_0 : baseline ROUGE-L variance > PIRC ROUGE-L variance. Significance level $\alpha = 0.01$.
- **Wilcoxon Signed-Rank (Quality):** Two-sided test — verifies PIRC does not degrade mean ROUGE-L quality. Significance level $\alpha = 0.01$.

The non-parametric Wilcoxon test is used because ROUGE-L distributions are not guaranteed to be normal, especially on small article samples.

6.3 Bootstrap Confidence Intervals

95% Bootstrap CI on mean paired difference (PIRC – Baseline)

- 10,000 resamples with replacement, seed = 42 for reproducibility.
- Applied to both variance reduction and ROUGE-L quality change.

6.4 Effect Sizes

Cohen's d_z = $\text{mean}(\text{diff}) / \text{std}(\text{diff})$ [paired, per-article differences]

- Computed for both ROUGE-L variance reduction and ROUGE-L mean quality change.
- Interpretation: $|d_z| < 0.2$ = negligible, $0.2-0.5$ = small, $0.5-0.8$ = medium, > 0.8 = large.

6.5 Sensitive Layer Analysis

- $\blacksquare^*$ distribution statistics: mean, std, min, max, coefficient of variation across articles.
- $S(\blacksquare)$ sensitivity curves plotted for 5 sample articles.
- Distribution analysis validates that $\blacksquare^*$ is consistently identifiable across the dataset.

6.6 Deterministic Reproducibility

- Global seed: Python random, NumPy, PyTorch CPU/CUDA seeded to 42 at pipeline entry.
- cuDNN determinism: `torch.use_deterministic_algorithms(warn_only=True)`, `PYTHONHASHSEED=42`.

- Greedy decoding (`do_sample=False`) eliminates sampling noise from cross-prompt output differences.

7. Key Formulas Reference

Component / Score	Formula / Value
PRI (primary ranking)	$0.40 \times \text{SMS} + 0.35 \times \text{CS} + 0.25 \times \text{Faithfulness}$
Faithfulness	$\text{mean}_k [P(\text{entail} \text{src} \rightarrow \text{resp}_k) + 0.5 \cdot P(\text{neutral} \dots)]$
Final Score	$0.60 \times \text{PRI} + 0.40 \times \text{Human_Score}$
SMS	$\text{mean}(\text{cosine_sim}) - 0.5 \times \text{Var}(\text{L2-normed embeddings})$
TRD	$\text{Var}(\text{embedding distances})$ across variants [lower = better]
KPIG	$\text{mean}_k [\text{facts}_k \cap \text{all_facts} / \text{all_facts}]$
CS	$0.40 \times \text{semantic} + 0.20 \times \text{length_adequacy} + 0.40 \times \text{coverage}$
ORI (cross-model ranking)	$(\text{SMS} + \text{AUC-E} + (1 - \text{TRD}) + \text{KPIG}) / 4$
IFI (intra-model diag.)	$1 - (\text{PPL_var} + \text{BF}) / 2$
Diagnostic_ORI	$\text{WHM}(\text{SMS}, \text{AUC-E}, \text{KPIG}, 1 - \text{TRD}) \text{ w}=0.40/0.30/0.20/0.10$
Diagnostic_IFI	$\text{WHM}(1 - \text{PPL_var}, 1 - \text{BF} [, 1 - \text{PC_stab}]) \text{ w}=0.50/0.30/0.20$
Diagnostic_PRI	$\text{HM}(\text{Diagnostic_ORI}, \text{Diagnostic_IFI}) \text{ w}=0.50/0.50$
$S(\blacksquare)$ sensitivity	$\text{Var}_k [\text{mean_token_PPL at layer } \blacksquare]$
$\blacksquare^*$ sensitive layer	$\text{argmax}_{\blacksquare} [S(\blacksquare) - S(\blacksquare-1)]$
Anchor score_t	$\text{rank}(\text{mean_ppl_t}) + \text{rank}(\text{var_ppl_t}) \rightarrow \text{bottom } 30\%$
PIRC clamping	$h[\text{anchors}] = \alpha \times \text{mean_h} + (1 - \alpha) \times h_{\text{original}}$
Variance Reduction	$1 - \text{mean}(\text{var_pirc}) / \text{mean}(\text{var_baseline})$
Cohen's dz	$\text{mean}(\text{diff}) / \text{std}(\text{diff})$

8. Key Thresholds & Constants

Parameter	Value	Location
SBERT similarity gate (lower)	≥ 0.82	paraphrase_generator.py
SBERT similarity gate (upper)	≤ 0.98	paraphrase_generator.py
SBERT threshold — dialogue task	≥ 0.78	paraphrase_generator.py
Token Jaccard overlap ceiling	≤ 0.85	paraphrase_generator.py
NLI bidirectional entailment (QA/Dial.)	≥ 0.50	paraphrase_generator.py
Max variants per strategy	2	paraphrase_generator.py
Paraphrase variants per instance (K)	8	config.yaml
Hallucination penalty threshold	$HS > 0.5 \rightarrow PRI \times 0.6$	evaluator.py
Short output penalty	$avg_len < 12 \rightarrow PRI \times 0.85$	scores.py
Entity coverage penalty	$coverage < 0.20 \rightarrow CS \times 0.60$	correctness_metric.py
CS composition	$0.40 \cdot sem + 0.20 \cdot len + 0.40 \cdot cov$	correctness_metric.py
Anchor percentile (bottom)	30%	anchor_tokens.py
Layer scan start fraction	25% of depth	sensitive_layer.py
Z-score fallback threshold	2.0	sensitive_layer.py
PIRC default α (clamping strength)	1.0 (full replacement)	config.yaml
Wilcoxon significance level	$\alpha = 0.01$	evaluate.py
Bootstrap resamples	10,000 (seed=42)	evaluate.py
Diagnosis dual-pillar threshold	0.70 (both ORI and IFI)	scores.py

Diagnostic ORI weights	SMS=0.40/AUC-E=0.30/KPIG=0.20/TRD=0.10	scores.py
Diagnostic IFI weights	PPL_var=0.50/BF=0.30/PC_stab=0.20	scores.py
PRI weights	Consistency=0.40/CS=0.35/Faith.=0.25	scores.py
Final Score weights	PRI=0.60/Human=0.40	scores.py
Dynamic weighting cap	trust_human ≤ 0.70	evaluator.py
max_new_tokens (generation)	200	config.yaml

9. Model Configuration

Role	Model	Quantization
Paraphrase Generator	meta-llama/Llama-3.1-70B-Instruct-AWQ-INT4	AWQ Marlin INT4
Subject #1	meta-llama/Llama-3.1-8B-Instruct	bf16
Subject #2	Qwen/Qwen2.5-7B-Instruct	bf16
Subject #3	mistralai/Mistral-7B-Instruct-v0.3	bf16
Subject #4 / 70B	hugging-quants/Meta-Llama-3.1-70B-Instruct-AWQ-INT4	AWQ Marlin INT4
LLM Judge	hugging-quants/Meta-Llama-3.1-70B-Instruct-AWQ-INT4	AWQ Marlin INT4
Faithfulness NLI	(same as judge — LLM-as-NLI path)	AWQ Marlin INT4
Embedder (SMS/CS/TRD/KPIG)	BAAI/bge-large-en-v1.5	fp32 / bf16
SBERT Filter (GenSens)	sentence-transformers/all-mpnet-base-v2	fp32
PIRC Target	meta-llama/Llama-3.1-8B-Instruct (HF hooks)	fp16 device_map=auto

GPU Memory Strategy. The 80 GB A100/H100 card holds only one large (70B) model at a time. The pipeline uses a **generate-only** → **score-from-CSV** split: each subject model generates responses independently (no judge loaded), then the judge/NLI/embedder score from the saved responses.csv in a separate process. This prevents OOM and eliminates self-evaluation bias.

10. Methodology Evolution & Key Design Decisions

The methodology has undergone systematic refinement through six phases of audit and fix cycles. The following summarises the most significant design decisions and the empirical evidence that motivated them.

10.1 De-collinearising PRI (R3 — most significant change)

The original PRI formula ($\text{Consistency} + \text{CS} + \exp(-\text{HS})$) was empirically found to have $\text{SMS} \blacksquare \text{PRI}$ Pearson $r = 0.949$ — PRI was effectively a rescaled SMS. Three 'metrics' measured the same underlying signal. The fix introduced **NLI-based Faithfulness** as a genuinely independent third axis, replacing SMS-Wasserstein ($r \approx 0.98$ with SMS) and cosine-KPIG (also collinear with SMS).

10.2 Prefill-Only PIRC Clamping (Critical Bug Fix)

An early PIRC implementation applied the forward hook on every model forward call, including each single-token decode step. This overwrote newly generated token hidden states with the first anchor's consensus, collapsing autoregressive generation into degenerate repetition ('Here://://://...'). All previous PIRC results before this fix were invalid. The corrected implementation uses a **clamp_state latch** that disables the hook after the prefill pass.

10.3 Percentile Anchor Selection (Robustness Fix)

The original absolute-threshold anchor selection ($\text{mean_ppl} < \tau$ AND $\text{var_ppl} < \tau_{\text{var}}$) was architecture-dependent: GPT-2 baselines operate at PPL ~ 50 – 100 while LLaMA operates at PPL ~ 5 – 15 . A poorly tuned τ silently produced zero anchors, making PIRC a no-op with no warning. The percentile-based method is self-calibrating and always returns at least one anchor.

10.4 IFI Saturation and Normalisation

PPL_{var} and BF computed on raw model logits differed by orders of magnitude across seq2seq (encoder–decoder) and causal architectures, causing IFI to saturate at 1.0 for some models. This collapsed $\text{Diagnostic_PRI} = \text{HM}(\text{ORI}, \text{IFI})$ into Diagnostic_ORI . The fix normalises PPL_{var} and BF per-model (min-max) before computing IFI.

10.5 TRD Demotion in Diagnostic_ORI

TRD and SMS show Pearson $r = -0.98$ (almost perfectly collinear) in the 5-instance pilot. Equal weighting in the harmonic mean let a TRD-only failure ($\text{SMS}/\text{AUC-E}/\text{KPIG}=0.85$, $\text{TRD}=0.9$) drop the composite to ~ 0.30 even though three of four axes agreed. The empirically motivated weights ($\text{SMS}=0.40$, $\text{AUC-E}=0.30$, $\text{KPIG}=0.20$, $\text{TRD}=0.10$) demote TRD to a tie-breaker while preserving the harmonic-mean 'any zero $\rightarrow$ zero' diagnostic property.

10.6 GenSens Pipeline Connections (R1)

Phase 1 generated SBERT-filtered paraphrase variants, but Phase 2 discarded them and re-wrapped articles in 9 near-synonymous templates. Phases 1 and 2 were completely disconnected — prompt sensitivity was never measured on the generated paraphrases. R1 wired the GenSens output directly into the evaluator, making the benchmark measure what it claims to measure.

Full Changelog. A complete record of every methodology change — including old values, new values, reasoning, and re-run requirements — is maintained in **method.md** at the repository root (newest entries first). This document represents the current state of the methodology as of **June 04, 2026**.

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